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Re: BioChips R Coming

This article appeared on Monday, 5/6/96 on the Chicago Tribune's front page, written by staff writer Jon Van...

"A tiny chip implanted inside the human body to send and receive radio messages, long a popular delusion among paranoids, is likely to be marketed as a consumer item early in the next century. Several technologies already available or under development will enable electronics firms to make implantable ID locators, say futurists, and our yearning for convenience and security makes them almost irresistible to marketers." "This is currently very hot", said Edward Cornish, president of the World Future Society, based in Bethesda, Md. "The field is developing because the technology is becoming available to do it".

He added: "it's appeal will depend on what features are offered and the price. I'm sure a large number of people would want such products."

Inevitably, implantable radio locators conjure up visions of Big Brother and unscrupulous scientists abusing such technology to control the masses. But the researchers laying the foundations for this technology see their work as helping humankind, not subverting privacy.

They seek to aid people using wireless phones to summon emergency help, to track soldiers who become lost on maneuvers and to enable people to

get along without carrying cash by automatically crediting an account.

Animal

advocates already urge pet owners to have tiny identification chips implanted

in

their dogs and cats so if they are lost, shelters can identify them through a

national computerized database. The notion of using implantable chips to

control humans

isn't entirely absent, even in these early stages of the technologies development. Cornish noted that authorities have experimented for years

with fitting convicts with electronic monitors to allow them to leave jails

for limited reasons, such as work release. "The problem is that monitors

worn

outside the body can be tampered

with," Cornish said. "Implanted locators would be more difficult to get at.

You might see this used as a condition for parole." Several systems already are in place with the potential to locate people using radio signals. The most obvious, called GPS, for global positioning satellites, was launched by the military years ago and has become available for civilian applications. It uses satellites to map a person's position with great precision.

Some automobiles come equipped with GPS gadgets that can give drivers their location, and boaters use similar technology. Researchers want to combine such locators with equipment that monitors a person's health. Engineers in Salt Lake City have designed a device intended to determine whether someone wearing it is becoming too cold or too hot, a sign of exposure. "We want to highlight people who need attention early, when there is still time to get to them with help", said Peter Kind, a senior vice president at Sarcos Research Corp., which has developed a prototype GPS-based device that will be ready for field

tests this year. Sarco's initial target is the military. Civilian markets might include ill people who ususally would be restricted to nursing environments, Kind said. Right now the prototype equipment is worn on a belt, but the goal is to minaturize it into a chip. The monitors could be worn in the area of the ear canal or elsewhere in the body. Another means to track people relies upon the existing

network of cellular-phone transmitters. The cellular industry and emergency-response officials have proposed standards to the Federal Communications Commission that would enable police, fire, and ambulance dispatchers to find people who dial 911 from wireless phones. At present nearly one-quarter of

the
911 emergency calls made in the US come from wireless phones, and
half the
time the callers don't know their location, posing a major problems
for
emergency personnel. Developing computer systems to track locations
of so
many calls is a
daunting task, but it is consistent with the phone industry's goal
of one
day
assigning phone numbers to human beings, rather than equipment. Once
the
phone network becomes sophisticated enough to do this, it will smooth
the
way

for widespread monitoring of people's whereabouts. Companies already
market pagers for children so parents can keep in touch when
youngsters are
away from home. Adding the ability to pinpoint location at any time
is a
natural extension: keeping track of the child through a chip
implanted under
the skin
may be another. "People accept that increased communications makes
life
more
convenient at the same time it means there's no hiding place
anymore",
Bernard Beck, a Northwestern University sociologist says. "If I have
a
universal
ID implanted, I can cash a check anywhere in the world. There's no
worry
about credit cards being stolen. These are attractive matters".
Although
older
people might recoil at the notion of sticking gizmos inside
themselves,
younger
ones tend to like the idea, Beck said. "In the last generation
there's been a

radical change about surgery and altering your body. The resistance
to
having

tattoos and wearing body rings is dropping. Altering your appearance
through
surgery is no longer a cause for amazement." Problems may arise when
some
people decide that their implant makes them itch or when it
malfunctions and
doesn't send out a strong enough radio signal to be received. Beck
sees
another
potential problem: "You'll see people ripping off implants,
counterfeiting
them

subverting the technology just as they've done to past technologies".
"It's
common in a certain genre of science fiction for people to walk into
a place,

pick up any item they like and walk out with it, not having to pay
because
their

action automatically generates a debit to their account", said Dan
Polsby, a
Northwestern University law professor. "That would make for a very
open
society.

Implanting tracers in criminals could reduce incarceration because it
would
allow
them to be tracked at all times. And, he added, people wearing
locators
would
be deterred from committing crimes because of the likelihood they
would be
caught.

"I don't know that we've wrapped our minds around being
accountable
minute-by-minute. The legal implications of who owns this information
are
major", Polsby added. "It's one thing to have my hospital monitoring
my

heartbeat for
fibrillation, but it's an entirely different matter to have the
government
monitoring

my whereabouts." Although potential problems are huge, locator ID
chips

may
be inevitable, said Cornish of the World Future Society. Just as many
people
now

allow supermarket chains to keep computerized records of their
individual
purchases in return for price discounts, many will embrace the chips
for

the security and convenience they offer, Cornish said. "We all want
to walk
down

the street feeling safe", he said. "This technology offers that
promise along
with

the dilemma of lost privacy". Cornish believes, at least initially,
that such
chips

would be voluntary. But he acknowledges that "things that are
voluntary
today

have a way of becoming compulsory tomorrow". Cornish said he sees a
similar

attitude already regarding e-mail addresses and pagers. "If you tell
people
you

don't have an e-mail address, they ask, 'How can we contact you?'

Some

~ employers now require staff to wear pagers, to be locatable. Someday,
they
may require chips."
